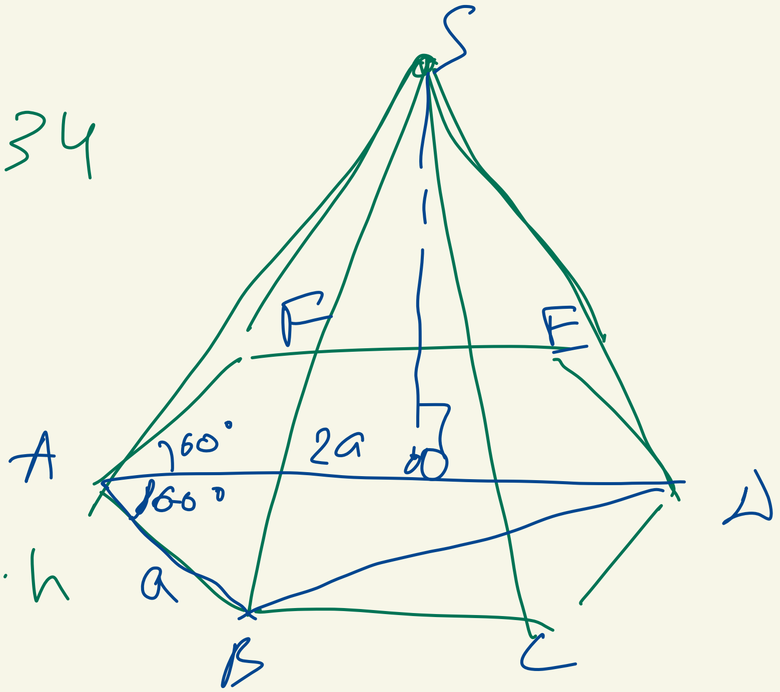
$S_{ABD} = 34$



$V = \frac{1}{3} S_{осн} \cdot h$

$$S_{ADB} = \frac{a \cdot 2a \cdot \sin 60°}{2} = \frac{a^2\sqrt{3}}{2}$$

$$S_{ABCDEF} = \frac{3a^2\sqrt{3}}{2}$$

$$V_{SABD} = \frac{1}{3} \frac{a^2\sqrt{3}}{2} \cdot SO$$

$$V_{SABCDEF} = \frac{1}{3} \frac{3a^2\sqrt{3}}{2} \cdot SO$$

$$\frac{V_{SABD}}{V_{SABCDEF}} = \frac{\cancel{\frac{1}{3}} \cancel{\frac{a^2\sqrt{3}}{2}} \cancel{SO}}{\cancel{\frac{1}{3}} \frac{3\cancel{a^2\sqrt{3}}}{\cancel{2}} \cancel{SO}} = \frac{1}{3}$$

$$V_{SABCDEF} = 3 \cdot 34 = 102$$

$$\operatorname{ctg} x - \sin x - \sqrt{3}\cos x + \frac{1}{\sin x} = 0$$

$$\frac{\cos x}{\sin x} - \sin x - \sqrt{3}\cos x + \frac{1}{\sin x} = 0$$

$$\cos x - \sin^2 x - \sqrt{3}\cos x \cdot \sin x + 1 = 0$$

$$\frac{\cos x + \cos^2 x - \sqrt{3}\cos x \sin x}{\sin x} = 0$$

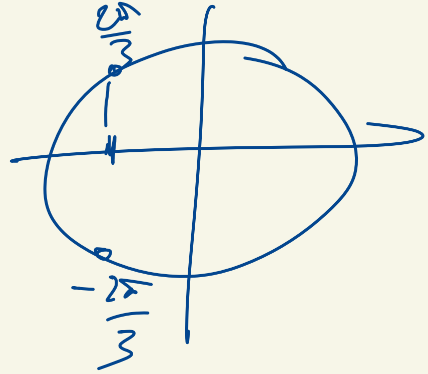
$$\frac{\cos x\,(1 + \cos x - \sqrt{3}\sin x)}{\sin x} = 0$$

$$\left[\begin{array}{l} \cos x = 0 \\ \cos x - \sqrt{3}\sin x = -1 \end{array}\right.$$

$$\frac{1}{2}\cos x - \frac{\sqrt{3}}{2}\sin x = -\frac{1}{2}$$

$$\cos\left(x + \frac{\pi}{3}\right) = \cos x \underbrace{\cos\frac{\pi}{3}}_{=\frac{1}{2}} - \sin x \underbrace{\sin\frac{\pi}{3}}_{=\frac{\sqrt{3}}{2}}$$

$$\left\{\begin{array}{l} \left[\begin{array}{l} \cos x = 0 \\ \cos\left(x + \frac{\pi}{3}\right) = -\frac{1}{2} \end{array}\right. \\ \sin x \neq 0 \end{array}\right.$$



$$\begin{cases} \left[\begin{array}{l} x = \frac{\pi}{2} + \pi k \\ x + \frac{\pi}{3} = \pm\frac{2\pi}{3} + 2\pi k \end{array}\right. \\ x \neq \pi k \end{cases}$$

$$\begin{cases} \left[\begin{array}{l} x = \frac{\pi}{2} + \pi k \\ x = -\frac{\pi}{3} \pm \frac{2\pi}{3} + 2\pi k \end{array}\right. \\ x \neq \pi k \end{cases}$$

$$\cancel{x = -\pi + 2\pi k}$$

$$x = \frac{\pi}{3} + 2\pi k$$

$$x = \frac{\pi}{2} + \pi k$$

$$x = \frac{\pi}{3} + 2\pi k$$

$$\left[\begin{array}{l} \sin x = \frac{\sqrt{3}}{2} \\ \sin x = -\frac{\sqrt{3}}{2} \end{array}\right. \leftarrow \sin^2 x = \frac{3}{4}$$

$$\underline{\sqrt{1 - \sin^2 x} = |\cos x|}$$

$$36^{-\cos x} - 36^{\cos x} = \frac{35}{6}$$

$$36^{\cos x} = t \qquad \frac{1}{t} - t = \frac{35}{6}$$

$$36^{-\cos x} = \frac{1}{t} \qquad \frac{6 - t^2 \cdot 6 - 35t}{6t} = 0$$

$$\frac{-6t^2 - 35t + 6}{6t} = 0$$

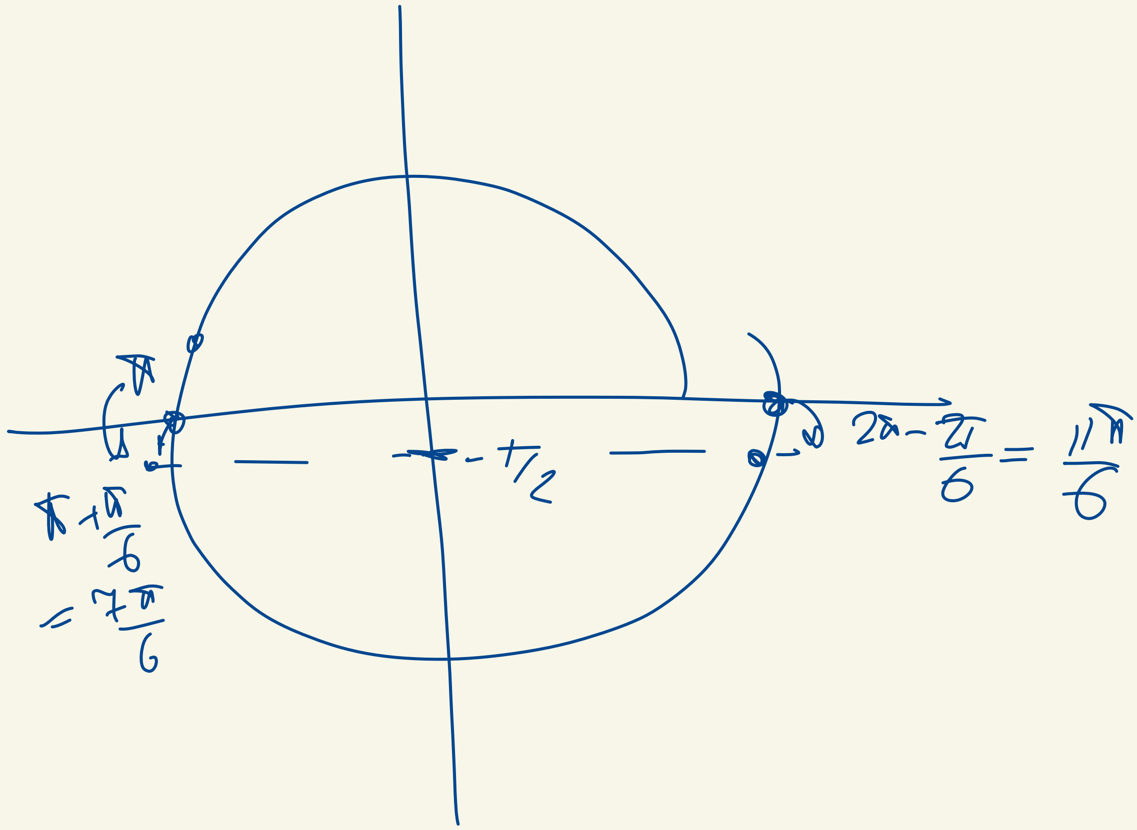
$$t \neq 0 \qquad \frac{35 \pm \sqrt{35^2 + 36 \cdot 4}}{12}$$

$$1225 + 144$$

$$\frac{35 \pm 37}{12} \to 6, \quad \to \frac{-2}{12} = -\frac{1}{6}$$

$$\left[\begin{array}{l} 36^{\cos x} = 6 \\ 36^{\cos x} = -\frac{1}{6} \end{array}\right. \to \left[\begin{array}{l} 2\cos x = 1 \\ \cancel{2\cos x} \; \varnothing \end{array}\right.$$

$$\cos x = \frac{1}{2}$$



$$\frac{7\pi}{6} + 2\pi k$$

$$\frac{11\pi}{6} + 2\pi k$$

$$7\sqrt{3}\sin x - \cos 2x - 10 = 1$$

$$7\sqrt{3}\sin x - (1 - 2\sin^2 x) - 11 = 0$$

$$\cos 2x = \underbrace{\cos^2 x}_{= 1 - \sin^2 x} - \sin^2 x$$

$$\cos^2 x + \sin^2 x = 1$$

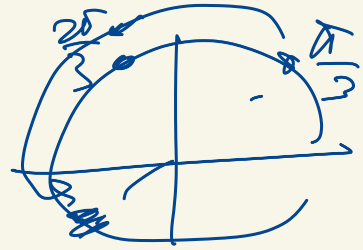
$$\cos^2 x = 1 - \sin^2 x$$

$$2\sin^2 x + 7\sqrt{3}\sin x - 12 = 0$$

$$2\cos^2 x = 2(1-\sin^2 x)$$

$$\frac{-7\sqrt{3} \pm \sqrt{(7\sqrt{3})^2 + 12\cdot 4\cdot 2}}{4}$$

$$\sin x = \frac{\sqrt{3}}{2} \rightarrow$$



$$\sin x = -4\sqrt{3} \quad \cancel{\varnothing}$$

$$\frac{\pi}{3} + 2\pi k$$

$$\frac{2\pi}{3} + 2\pi k$$

$$2\operatorname{tg}\left(\frac{3\pi}{2} - x\right)$$

$$\sin(3\pi - 2x) = \underbrace{\cancel{\sin 3\pi}}_{0}\cos x - \sin 2x \underbrace{\cancel{\cos 3\pi}}_{-1}$$

$$= \sin 2x$$

$$\operatorname{tg}\left(\frac{3\pi}{2} - x\right) = \frac{\sin\left(\frac{3\pi}{2} - x\right)}{\cos\left(\frac{3\pi}{2} - x\right)} = \frac{\overset{-1}{\cancel{\sin\frac{3\pi}{2}}}\cos x - \sin x\overset{0}{\cancel{\cos\frac{3\pi}{2}}}}{\underbrace{\cancel{\cos\frac{3\pi}{2}}}_{0}\cos x + \underbrace{\cancel{\sin\frac{3\pi}{2}}}_{-1}\sin x}$$

$$= \frac{-\cos x}{-\sin x} = \operatorname{ctg} x$$

$$\sqrt{2\operatorname{ctg} x \cdot \sin 2x} = -\operatorname{tg}\frac{2\pi}{3}$$

$$\frac{\sin\frac{2\pi}{3}}{\cos\frac{2\pi}{3}} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$$

$$\sqrt{2\operatorname{ctg} x \sin 2x} = \sqrt{3}$$

$$\begin{cases} 2\operatorname{ctg} x \sin 2x = 3 \\ \operatorname{ctg} x \sin 2x \geq 0 \end{cases}$$

$$2\frac{\cos x}{\cancel{\sin x}} \cos x \cancel{\sin x} \geq 0$$

$$\cos^2 x \geq 0$$

всегда

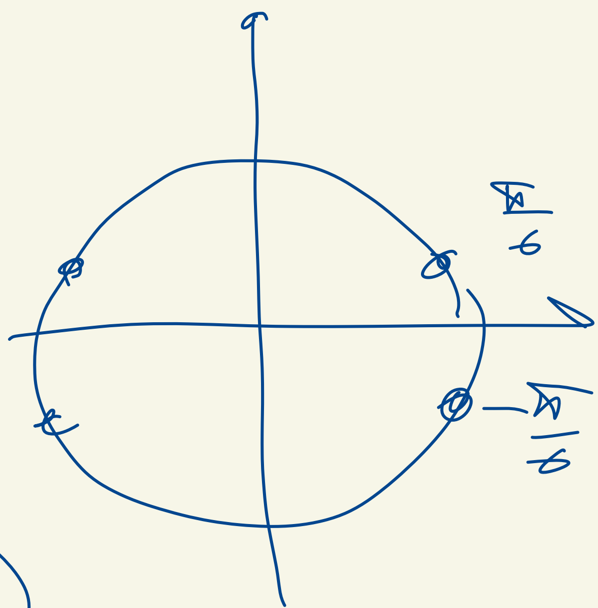
$$2 \cdot 2 \cancel{\sin x} \cos x \cdot \frac{\cos x}{\cancel{\sin x}} = 3$$

$\sin x \neq 0$

$$2 \cdot 2\cos^2 x = 3$$

$$\cos^2 x = \frac{3}{4}$$

$$\cos x = \pm\frac{\sqrt{3}}{2}$$



$$x = \pm\frac{\pi}{6} + \pi k$$

$$\frac{3\left(t^3-1\right)}{t-1}+\frac{20}{t^2+t+1}\le 19$$

$$t^3-1=(t-1)\left(t^2+t+1\right)$$

$$a^3-b^3=(a-b)\left(a^2+ab+b^2\right)$$

$$\underset{t}{\overset{\shortparallel}{a}} \qquad \overset{\shortparallel}{b}$$

$$8^x-1=2^{3x}-1^3=\left(2^x-1\right)\left(2^{2x}+2^x+1\right)$$

$$\frac{3\left(2^x-1\right)\left(2^{2x}+2^x+1\right)}{\left(2^x-1\right)}+\frac{20}{2^{2x}+2^x+1}\le 19$$

$$2^x-1\ne 0$$

$$3\left(2^{2x}+2^x+1\right)+\frac{20}{2^{2x}+2^x+1}\le 19$$

$$2^{2x}+2^x+1=t$$

$$t\ge \frac{4}{3}$$

$$3t+\frac{20}{t}\le 19$$

$$t\le 5$$

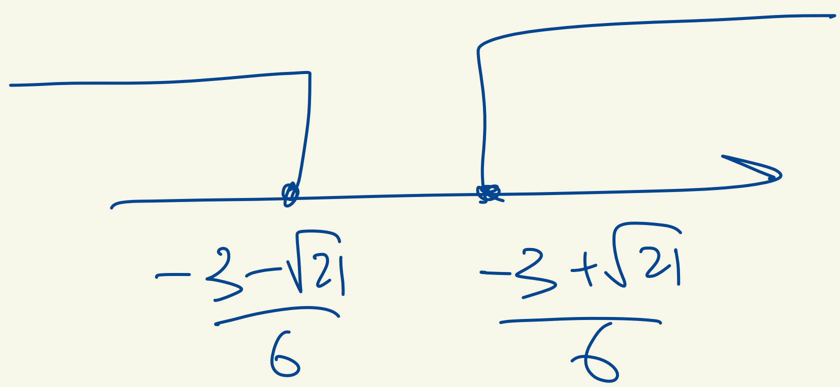
$$\begin{cases} 2^{2x}+2^{x}+1 \geq \frac{4}{3} \\ 2^{2x}+2^{x}+1 \leq 5 \end{cases}$$

$$\begin{cases} 3y^2+3y-\frac{1}{3} \geq 0 \\ y^2+y-4 \leq 0 \end{cases}$$

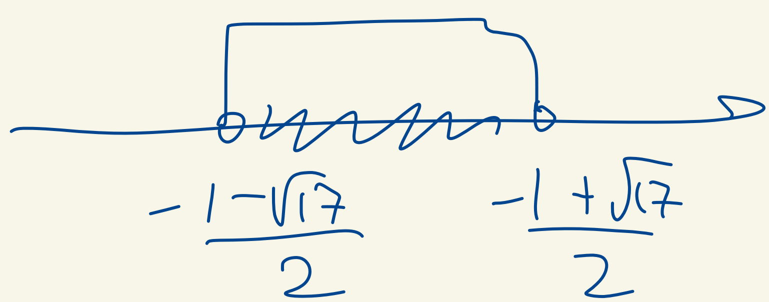
$9+12=21$

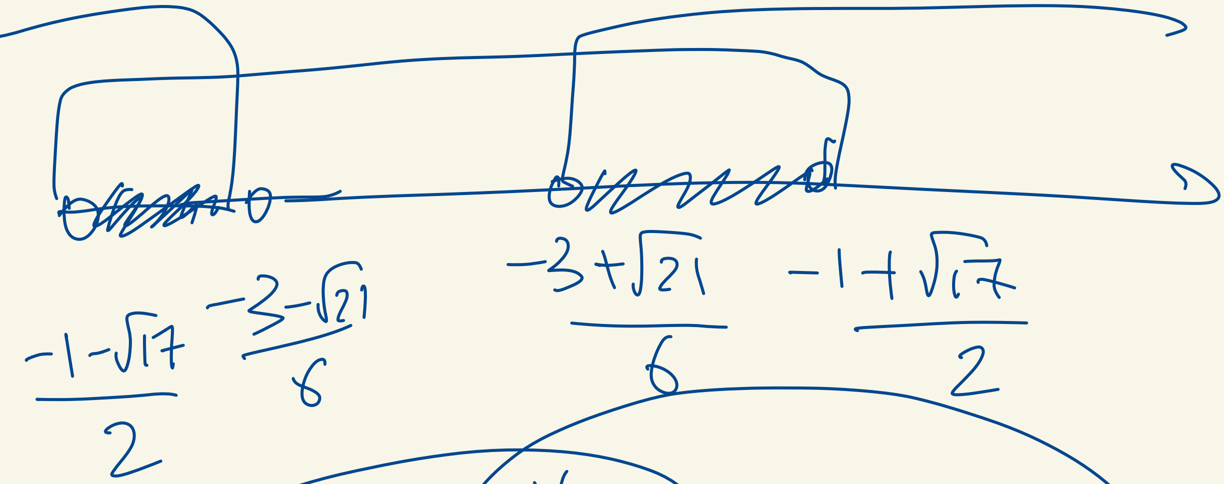
$$\frac{-3 \pm \sqrt{9+3\cdot 4}}{6}$$

$$\frac{-3 \pm \sqrt{21}}{6}$$



$$\frac{-1 \pm \sqrt{1+16}}{2} = \frac{-1 \pm \sqrt{17}}{2}$$





$$\frac{-1-\sqrt{17}}{2} \le 2^x \le \frac{-3-\sqrt{21}}{6} \qquad \varnothing$$

$$\frac{-3+\sqrt{21}}{6} \le 2^x \le \frac{-1+\sqrt{17}}{2}$$

$$\begin{cases} \log_2\left(\frac{3+\sqrt{21}}{6}\right) \le x \le \log_2\left(\frac{-1+\sqrt{17}}{2}\right) \\ 2^x \ne 1 \qquad x \ne 0 \end{cases}$$

$$x \in \left[\log_2\left(\frac{3+\sqrt{21}}{6}\right); 0\right) \cup \left(0; \log_2\left(\frac{\sqrt{17}-1}{2}\right)\right]$$

$$x^2 + (1-\sqrt{10})x - \sqrt{10} \le 0$$

$$\sqrt{2\sqrt{10}+11}$$

$$-(1-\sqrt{10}) \pm \sqrt{(1-\sqrt{10})^2 + 4\sqrt{10}}$$

$$1+10-2\sqrt{10}$$

$$\underline{\underline{11+2\sqrt{10}}}$$

$$1+2\sqrt{10}+10$$

$$\sqrt{(\sqrt{10}+1)^2} = 1+\sqrt{10}$$

$$\frac{-(1-\sqrt{10}) \pm (1+\sqrt{10})}{2}$$

$$\frac{-1+\sqrt{10}+1+\sqrt{10}}{2} = \sqrt{10}$$

$$\frac{-1+\sqrt{10}-1-\sqrt{10}}{2} = -1$$

$$[-1;\sqrt{10}]$$

$$x^2 - (3+\sqrt{5})x + 3\sqrt{5} \le 0$$

$$[-3-\sqrt{5} ; \sqrt{5}]$$

$$\Delta = 3-\sqrt{5}$$

$$\frac{3+\sqrt{5} \pm (3-\sqrt{5})}{2} = \quad 3^-$$

$$x\sqrt{2} > 0 \qquad \frac{x}{1-x} > 0$$

$$2\log_7 (x\sqrt{2}) - \log_7 \left(\frac{x}{1-x}\right) \le$$

$$\log_7 \left(8x^2 + \frac{1}{x} - 5\right)$$

$$\log_7 2x^2 - \log_7 \frac{x}{1-x} \le \log_7 8x^2 + \frac{1}{x} - 5$$

$$\log_7 \frac{2x^2}{x}(1-x) \le \log_7 8x^2 + \frac{1}{x} - 5$$

$$2x(1-x) \leq 8x^2 + \frac{1}{x} - 5$$

$$2$$

$$\frac{-10x^3 + 2x^2 + 5x - 1}{x} \leq 0$$

$$2x^2(-5x+1) + 5x - 1$$

$$(5x-1)(-2x^2+1)$$

$$\frac{(5x-1)(1-2x^2)}{x} \leq 0$$